**Université Catholique d'Afrique Centrale (UCAC)**

**Institut Catholique de Yaoundé (ICY)**

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| **COURSE TITLE:** | **MATHEMATICAL ANALYSIS** |
| **DEPARTMENTS:** | **BEM1** |
| **LECTURER:** | **ARREY Marinus** |
| **INSTRUCTIONS:** | **ANSWER ALL QUESTIONS** |
| | **SECOND SEMESTER EXAMINATION June 2026** |

**Question 1: (4marks)**

Find the critical points of the following function. Determine whether the critical points correspond to a relative maximum or to a relative minimum or to the neither or the test gives no information.

a) $F(x,y) = -X^3 + XY + Y^2 + X$

b) $f(X) = X^3 - 6X^2 + 16$.

**Question 2: (4marks)**

Use algebra to solve the following equations:

a) $3^{2x} - 9(3^x) + 18 = 0$ **(2marks)**

b) $\text{Log}_2(2x) = \log_2(5x + 4) - 3$ **(2marks)**

**Question 3: (4marks)**

Suppose we want to optimize the objective function of a firm givens as;

$f(X_1, X_2) = 5X_1^2 + 6X_2^2 - X_1X_2$ ; subject to $X_1 + 2X_2 = 24$

**Question 4: (4marks)**

The production function of a commodity is: $Q = 10L - 0.1L^2 + 15K - 0.2K^2 + 2KL$ where L is labour, K is capital and Q is production.

i) Calculate the marginal products of the two inputs when 10 units of each labour and capital are used,

ii) If 10 units of capital are used, what is the upper limit for the use of labour which a rational producer will never exceeds.

iii) Determine the marginal rate of technical substitution (MRTS) for LK and for KL

**Question 5: (4marks)**

The demand equation for a product is; $P = (q - 5)^2$ and

the supply equation is; $P = P^2 + q + 3$; where P (in thousands of FCFA) is the price per unit and where q units are demanded or supplied. Determine the consumer surplus as well as the producer surplus.

**Good Luck!**